LOW VOLTAGE SHUNT CAPACITORS · DRY TYPE

PCMK

Self-healing dry-type power capacitors for low voltage power factor correction
2.5 – 50 kVAr · 400 · 440 · 480 · 525 V · IEC 60831

50 kVAr SINGLE UNIT UP TO	**525 V** RATED VOLTAGE	**PSD** OVERPRESSURE DISCONNECTOR	**IEC** 60831-1 / -2 COMPLIANT

01 · COMPANY PROFILE

ENGINEERING THE FUTURE OF POWER WITH PRECISION AND INNOVATION

Since our inception in 2012, Powerline has been driven by a singular mission: to serve as the reliable backbone of the regional and Egyptian power sector. We are not merely a contracting and supply firm — we are a fully integrated engineering partner specialising in the design and manufacture of Medium Voltage (MV) and Low Voltage (LV) power distribution systems, adhering to the most stringent international technical standards.

A Legacy of Reliability

Built on foundations of commitment and precision, Powerline has established itself as a hallmark of quality in major national and industrial projects. Our track record reflects a consistent ability to overcome complex engineering challenges and deliver customised solutions for the modern electrical grid.

Solutions Beyond Expectations

Our MV switchgear, Ring Main Units and LV main and sub-main distribution boards are characterised by exceptional operational efficiency and a long life cycle. We bridge mechanical durability with digital intelligence, ready for integration with the latest smart-grid and control technologies.

Quality is not a department — it is our work culture

Production & Manufacturing

Advanced production lines ensure the highest accuracy in the assembly and calibration of MV & LV switchgear, in full compliance with IEC international standards.

Engineering & Design

Our design engineers develop layouts that prioritise maximum safety, operational continuity and seamless serviceability across the installation's life.

Testing & Quality Control

Every unit leaving our facilities undergoes rigorous routine testing to ensure durability under the harshest operating conditions — total peace of mind for our clients' investments.

2012	2	MV+LV	IEC
ESTABLISHED · CAIRO	MANUFACTURING FACILITIES	INTEGRATED PORTFOLIO	CERTIFIED QUALITY SYSTEM

" Powerline: Performance Excellence, Execution Precision & Sustainable Solutions. "

PCMK · LOW VOLTAGE CAPACITORS

TABLE OF CONTENTS

How to use this catalogue

Sections 02–04 introduce the capacitor and what it does. Sections 05–08 carry the standards, the selection table, the output at each service voltage and the dimensions needed to specify a unit. Sections 09–12 give the detailed ratings, how the capacitor is built, and the protection, installation and quality evidence supplied with every delivery.

02 · PRODUCT OVERVIEW

PCMK AT A GLANCE

PCMK is a self-healing, dry-type low voltage shunt capacitor for power factor correction on 400 to 525 V three-phase networks. Metallised polypropylene windings are sealed in a cylindrical aluminium can with an integral overpressure disconnector — no oil, no gas, no impregnating liquid to leak or age.

- Three-phase, delta connected, 50 Hz — ready for direct connection to the busbar
- Rated 525 V; the same unit delivers a defined output at 480, 440 and 400 V
- Self-healing metallised polypropylene (MPP) dielectric — a puncture clears itself in microseconds
- Pressure-sensitive overpressure disconnector (PSD) isolates the unit at end of life
- Discharge resistors fitted as standard
- Dry construction — PCB free, no impregnating oil, safe to handle and dispose of
- Eleven ratings from 2.5 to 50 kVAr in one mechanical family

PCMK DRY-TYPE CAPACITOR — CYLINDRICAL ALUMINIUM CAN

2.5 – 50	525 V	Δ	-5/D
KVAR PER UNIT	RATED VOLTAGE	DELTA CONNECTION	TEMPERATURE CATEGORY

Designation

Units are ordered by item code: **PCMK · 0525 · kVAr · 3** — PCMK dry-type capacitor, 0525 rated voltage in volts, the output in kVAr at 525 V (R replaces the decimal point), 3 phases. **PCMK0525503** is therefore a 50 kVAr, 525 V, three-phase unit. The full list is on page 8.

03 · TECHNOLOGY

CAPACITOR TECHNOLOGY

PCMK uses a metallised polypropylene dielectric in a dry, cylindrical aluminium can. Three elements define its behaviour in service: self-healing, dry construction and the overpressure disconnector.

Self-healing dielectric

The polypropylene film carries a metallisation only a few nanometres thick. At a weak point the dielectric punctures, and the surrounding metallisation evaporates and clears the fault in microseconds. The capacitor keeps working with a negligible loss of capacitance.

Dry, PCB free

No impregnating oil or gas: nothing to leak, nothing to dispose of as hazardous waste, and no risk of fire being fed by the impregnant. The winding is sealed in an aluminium can with a granulate filling.

Overpressure disconnector

At end of life the accumulated self-healing gas raises the internal pressure. The can lid expands and tears the internal fuse links, disconnecting all three phases before the case can rupture (OD: PSD).

How the overpressure disconnector works

STAGE 1 · NORMAL

Windings connected through the internal links. The lid sits flat; the can is at ambient pressure.

STAGE 2 · AGEING

Repeated self-healing releases small volumes of gas. Pressure rises slowly over the service life and the lid begins to expand.

STAGE 3 · DISCONNECT

Expansion pulls the links apart. All three phases are isolated, the unit goes open-circuit and the bank simply loses one step.

Why self-healing matters

A non self-healing capacitor fails as a short circuit and must be cleared by the upstream device. A self-healing unit degrades gracefully: capacitance drifts slowly downward, and the overpressure disconnector removes the unit cleanly when it is spent.

Discharge is built in

Discharge resistors are fitted inside every unit. After disconnection the residual voltage falls to a safe value within one minute — the interval stated on the nameplate warning — without any external discharge equipment.

04 · FEATURES

PRODUCT FEATURES

Performance

- Self-healing MPP dielectric
- Low dielectric losses
- Stable capacitance over life
- Delta connected — no neutral required
- 50 Hz, three phase

Safety

- Overpressure disconnector on every unit
- Discharge resistors fitted as standard
- Dry construction — no oil to leak
- PCB free
- Insulation level 3 / 8 kV

Durability

- Sealed aluminium can
- Temperature category -5/D, up to 55 °C
- Tolerant of 1.3 × rated current
- Withstands cyclic switching duty
- Vertical or horizontal mounting

Cost saving

- One can family across 2.5–50 kVAr
- Compact footprint in the panel
- No maintenance beyond periodic checks
- Local manufacture — short lead times
- Graceful end of life, one step at a time

Rated at 525 V for a reason

A capacitor applied on a 400 V network sees more than 400 V in practice: supply tolerance, light-load voltage rise and the voltage across a detuned reactor all add up. Specifying the 525 V rating and accepting the reduced output at 400 V keeps the dielectric stress low and multiplies the service life. Output at each voltage is tabulated on page 9.

1.10 U_n

8 H PER 24 H

1.30 I_n

CONTINUOUS OVERCURRENT

05 · STANDARDS

STANDARDS & SERVICE CONDITIONS

PCMK capacitors are designed, manufactured and routine-tested to the IEC 60831 series for self-healing shunt power capacitors.

Standard	Scope
IEC 60831-1	Shunt power capacitors of the self-healing type for a.c. systems up to and including 1 kV — Part 1: General — performance, testing and rating; safety requirements; guide for installation and operation
IEC 60831-2	Part 2: Ageing test, self-healing test and destruction test
IEC 60871	Applied where capacitors are combined with detuned reactors in filter duty
IEC 61921	Power capacitors — low voltage power factor correction banks
IEC 60529	Degree of protection of the enclosure into which the capacitor is built

Service conditions

Condition	Specification
Minimum ambient temperature	-5 °C
Maximum ambient temperature	+55 °C
Mean over any 24 h	+45 °C
Mean over one year	+35 °C
Temperature category	-5/D
Installation altitude	Up to 2,000 m above sea level
Relative humidity	Non-condensing
Installation	Indoor, in a ventilated enclosure
Mounting position	Vertical or horizontal

Why it matters

- **Category -5/D** — the highest ambient class in IEC 60831-1. The unit is rated for the temperatures found inside a closed capacitor panel in a hot climate, not just for laboratory conditions.
- **Self-healing verified** — IEC 60831-2 requires ageing, self-healing and destruction tests. The destruction test proves the overpressure disconnector operates before the case ruptures.
- **Tolerance -5 % / +10 %** — capacitance is controlled at manufacture so that a bank reaches its declared output without over-compensating at light load.

Above 2,000 m, in condensing humidity or outside the -5 °C to +55 °C range, contact Powerline engineering — a derating study or a different voltage rating may be required.

06 · RATINGS

RATING & SELECTION TABLE

Eleven ratings, one mechanical family. Output and current are given at the rated voltage of 525 V; the corresponding figures at 480, 440 and 400 V are on page 9.

Item code	Specification	Output kVAr	Current A	Capacitance μF	D × H mm
PCMK05252R53	525-2.5-3	**2.5**	3	28.9	65 × 105
PCMK0525053	525-5-3	**5**	5	57.8	65 × 160
PCMK05257R53	525-7.5-3	**7.5**	8	86.7	65 × 240
PCMK0525103	525-10-3	**10**	11	115.5	76 × 240
PCMK052512R53	525-12.5-3	**12.5**	14	144.4	76 × 240
PCMK0525153	525-15-3	**15**	16	173.3	86 × 240
PCMK0525203	525-20-3	**20**	22	231.1	86 × 290
PCMK0525253	525-25-3	**25**	27	288.9	96 × 290
PCMK0525303	525-30-3	**30**	33	346.6	106 × 290
PCMK0525403	525-40-3	**40**	44	462.2	136 × 290
PCMK0525503	525-50-3	**50**	55	577.7	136 × 290

Values at 525 V, 50 Hz, delta connection. Capacitance is the total three-phase value measured at the terminals, tolerance -5 % / +10 %. Current is the rated line current $I_n = Q / (\sqrt{3} \cdot U_n)$; allow 1.3 × I_n when sizing cables, contactors and protective devices, and 1.43 × I_n where the capacitance tolerance is taken at its upper limit.

How to read the item code

PCMK 0525 50 3 — **PCMK** Powerline dry-type capacitor · **0525** rated voltage in volts · **50** output in kVAr at that voltage, with **R** in place of a decimal point (2R5 = 2.5 kVAr) · **3** three phase.

Building a bank

Steps are normally built from equal or binary-weighted units — for example 4 × 25 kVAr, or 12.5 / 25 / 25 / 50 kVAr for finer resolution at light load. Powerline sizes the step pattern, contactors and detuned reactors to the measured load profile.

07 · OUTPUT

OUTPUT AT OTHER VOLTAGES

A capacitor delivers reactive power in proportion to the square of the applied voltage. The same PCMK unit therefore gives a lower — and entirely predictable — output when it is applied below its 525 V rating. Always select on the output required at the **actual service voltage**.

100 %	83.6 %	70.2 %	58 %
AT 525 V	AT 480 V	AT 440 V	AT 400 V

Output factor = $(U_{service} / 525\ V)^2$, applied to the rated kVAr of the unit.

Item code	525 V		480 V		440 V		400 V	
	kVAr	A	kVAr	A	kVAr	A	kVAr	A
PCMK05252R53	2.5	3	2.1	3	1.8	2	1.5	2
PCMK0525053	5	5	4.2	5	3.5	5	2.9	4
PCMK05257R53	7.5	8	6.3	8	5.3	7	4.4	6
PCMK0525103	10	11	8.4	10	7	9	5.8	8
PCMK052512R53	12.5	14	10.4	13	8.8	12	7.3	10
PCMK0525153	15	16	12.5	15	10.5	14	8.7	13
PCMK0525203	20	22	16.7	20	14	18	11.6	17
PCMK0525253	25	27	20.9	25	17.6	23	14.5	21
PCMK0525303	30	33	25.1	30	21.1	28	17.4	25
PCMK0525403	40	44	33.4	40	28.1	37	23.2	34
PCMK0525503	50	55	41.8	50	35.1	46	29	42

All values at 50 Hz, delta connection, nominal capacitance. Currents are rounded to the nearest ampere and match the values marked on the unit nameplate.

Worked example

A 400 V board needs 100 kVAr of compensation. A 525 V unit gives 58.1 % of its rating at 400 V, so 100 / 0.581 = 172 kVAr of installed 525 V capacity is required — for example four PCMK0525503 units (4 × 29.0 = 116 kVAr) plus two PCMK0525303 (2 × 17.4 = 34.8 kVAr) and one PCMK0525503 spare step, or the nearest practical step pattern.

With detuned reactors

A detuned reactor raises the voltage across the capacitor above the network voltage — typically by 6 to 14 % depending on the tuning factor. The capacitor rating and the delivered output must both be recalculated for that raised voltage; Powerline engineering performs this calculation with the order.

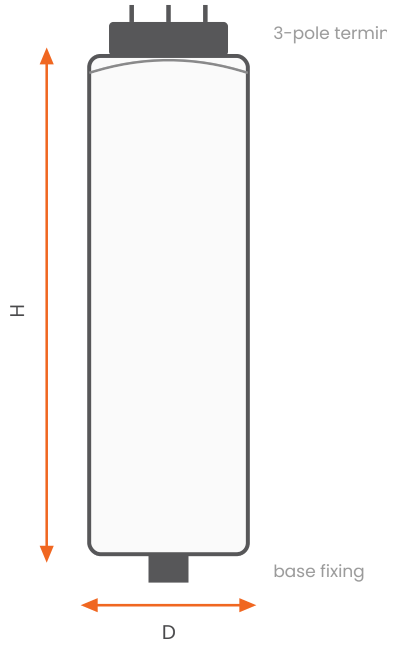
08 · DIMENSIONS

DIMENSIONS & MOUNTING

Four can diameters and three heights cover the whole range. D is the outside diameter of the aluminium can and H is the overall height of the body, measured from the base to the top of the lid.

D mm	H mm	Output kVAr	Item code
65	105	**2.5**	PCMK05252R53
65	160	**5**	PCMK0525053
65	240	**7.5**	PCMK05257R53
76	240	**10**	PCMK0525103
76	240	**12.5**	PCMK052512R53
86	240	**15**	PCMK0525153
86	290	**20**	PCMK0525203
96	290	**25**	PCMK0525253
106	290	**30**	PCMK0525303
136	290	**40**	PCMK0525403
136	290	**50**	PCMK0525503

Dimensions exclude the terminal block and the base fixing. Tolerance on D and H: ±2 mm.



OVERALL DIMENSIONS

Mounting

- Vertical or horizontal, secured through the base fixing
- Terminal block accessible for connection and inspection
- Leave free space around each can for convection — do not pack units against one another
- Enclosure ventilated so that the ambient at the capacitor stays within category -5/D
- Earth the can to the panel earth bar

09 · TECHNICAL DATA

MAIN TECHNICAL CHARACTERISTICS

Ratings common to the whole PCMK range. Output, current, capacitance and dimensions for each individual unit are given in the rating table on page 8.

Characteristic	Unit	PCMK 0525 series
Rated voltage (U_n)	V	525
Service voltages	V	400 · 440 · 480 · 525
Rated output at 525 V	kVAr	2.5 · 5 · 7.5 · 10 · 12.5 · 15 · 20 · 25 · 30 · 40 · 50
Rated frequency	Hz	50
Number of phases	—	3
Internal connection	—	Delta (Δ)
Capacitance tolerance	—	-5 % / +10 %
Insulation level (power frequency / impulse)	kV	3 / 8
Temperature category	—	-5/D
Permissible overvoltage	× U_n	1.10 · 8 h/24 h 1.15 · 30 min/24 h 1.20 · 5 min · 1.30 · 1 min
Permissible continuous overcurrent	× I_n	1.30
Maximum current with capacitance tolerance	× I_n	1.43
Dielectric	—	Metallised polypropylene, self-healing
Impregnation	—	Dry · PCB free
Enclosure	—	Cylindrical aluminium can
End-of-life protection	—	PSD overpressure disconnector
Discharge device	—	Internal discharge resistors
Mounting	—	Vertical or horizontal
Standards	—	IEC 60831-1 / IEC 60831-2

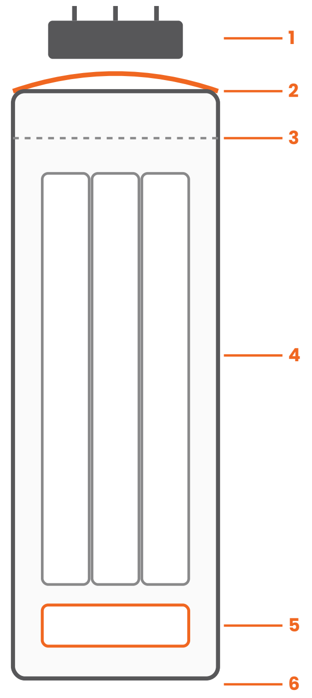
Overvoltage and overcurrent limits are the values permitted by IEC 60831-1 for this capacitor class; they are not intended to be applied simultaneously or continuously. Temperature category -5/D corresponds to a minimum ambient of -5 °C and a maximum of +55 °C.

10 · DESIGN

CONSTRUCTION & DESIGN

Each PCMK unit is a sealed cylindrical assembly. Three wound elements are connected in delta inside the can, held in a granulate filling that anchors the winding and conducts heat to the aluminium wall.

Part	Description
Terminal block	Three-pole insulated terminal block on the lid, protected against direct contact
Expansion lid	Aluminium lid with expansion grooves — the moving element of the overpressure disconnector
Internal links	Break-away connections between lid and windings; torn apart when the lid expands
Windings	Three self-healing metallised polypropylene elements, delta connected
Discharge resistors	Fitted across the terminals inside the can
Filling	Inert dry granulate — mechanical support and heat transfer
Can	Drawn aluminium cylinder, sealed; base fixing



SECTIONAL ARRANGEMENT

1	Terminal block
2	Expansion lid
3	Break-away links
4	MPP windings
5	Discharge resistors
6	Aluminium can & base fixing

Sealed for the life of the unit

There is nothing to top up, re-impregnate or re-tension. The only service actions are the periodic checks listed on page 13 — terminal tightness, capacitance measurement and a visual check of the lid for expansion.

11 · PROTECTION & INSTALLATION

PROTECTION, SWITCHING & INSTALLATION

A capacitor is a low-impedance load at switch-on and a source of stored energy at switch-off. Both have to be handled in the panel design.

Protection & switching

Item	Requirement
Cable and busbar sizing	Minimum 1.3 × I_n; 1.43 × I_n where the capacitance tolerance is taken at +10 %
Switching device	Contactor rated for capacitor duty (AC-6b) with pre-charging resistor contacts
Short-circuit protection	HRC fuse or circuit breaker sized above the continuous capacitor current and below the cable rating
Inrush limitation	Pre-charging resistors in the contactor, or a series choke, on every switched step
Reconnection delay	A step must not be re-energised until it has discharged — minimum one minute
Harmonic duty	Detuned reactors where the distortion or the capacitor-to-transformer ratio calls for them
Earthing	Can bonded to the panel earth bar

Discharge — the one minute rule

Internal discharge resistors bring the residual voltage down to a safe value within one minute of disconnection. Never touch the terminals, the discharge resistor or the overvoltage isolator before that minute has passed, and always verify with a voltage tester before working on a bank.

Installation

- Install indoors, in a ventilated enclosure, clear of direct heat sources
- Check that the nameplate voltage matches the service voltage before energising
- Tighten terminal connections to the terminal-block manufacturer's torque — loose joints are the commonest cause of failure
- Allow convection space between cans and above the terminal block
- Verify step sequence and contactor timing before handing over to automatic control

Periodic checks

- Visual check of the lid — an expanded lid means the unit has reached end of life and must be replaced
- Measure capacitance; replace a unit that has lost more than the permitted margin
- Re-torque terminals and confirm ventilation is unobstructed
- Confirm the ambient inside the panel is still within category -5/D

1.3 ×	AC-6b	1 min	PSD
I_N MINIMUM CABLE SIZING	CONTACTOR DUTY CLASS	DISCHARGE BEFORE TOUCHING	END-OF-LIFE DISCONNECTION

12 · QUALITY

NAMEPLATE, QUALITY & TESTING

Routine tests — every unit

- Capacitance measurement and verification against the declared tolerance
- Voltage test between terminals
- Voltage test between terminals and case, to the declared insulation level
- Sealing test on the can
- Discharge resistor continuity and discharge time verification
- Visual and dimensional inspection, marking check

Type verification

- Ageing test, self-healing test and destruction test to IEC 60831-2 — the destruction test proves the overpressure disconnector opens before the case can rupture
- Thermal stability at the upper limit of temperature category D
- Impulse and power-frequency withstand at the declared 3 / 8 kV insulation level

> **Documentation with every delivery**
>
> Routine test records, the declaration of conformity, the rating and dimension data for the supplied units and the installation and safety instructions are supplied with each consignment.

EVERY UNIT CARRIES A FULL RATING PLATE

What the nameplate tells you

Q / V / I	Output, voltage and current at each service voltage
MODEL	Powerline item code
CAPACITANCE	Total three-phase value in μF
CONNECTION	Delta (Δ)
INSULATION LEVEL	Power frequency / impulse withstand
CATEGORY	Temperature category to IEC 60831-1
OD	Overpressure disconnector type
PCB FREE	No polychlorinated biphenyls

بـــاورلايـــن
POWERLINE

Performance excellence.
Execution precision.
Sustainable solutions.

From engineering and manufacture to testing, delivery and lifetime support — Powerline is your integrated partner for MV & LV power distribution in Egypt and the region.

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For technical enquiries or quotations, please get in touch with our team — we respond with engineered solutions, not just prices.

2.5 – 50
KVAR PER UNIT

525 V
RATED VOLTAGE

IEC 60831
-1 / -2 COMPLIANT

PCB FREE
DRY TYPE

Data in this catalogue is provided for product presentation and is subject to change without notice; contract values are confirmed at order stage. Capacitor ratings are declared at 50 Hz and at the stated service voltage; final selection is confirmed by Powerline engineering against the site load profile.